

Mapping Health: Analyzing Provider Distribution and Government Investment Across U.S. States

CS 3891 Project Status Report
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Objective

Analyze the correlations between healthcare provider density and government healthcare investment across U.S. states.

Goals

- Explore regional and state disparities
- Identify key states where gaps in healthcare resources might exist
- Finding correlations between these with state-level healthcare spending.

Materials + Description of Data

ONC = Office of the National
Coordinator for Health Information
Technology

Datasets

- **Provider Data**
 - SK&A Database from ONC (2011-2013) on counts of various types of healthcare providers – 153
- **State Govt. Investment Data**
 - Urban Institute's dataset (2011-2013) from U.S. Census Bureau financial data on healthcare expenditure by state (~200)

Sample Provider Data →→

Sample Investment Data →→

```
3  {
4    "region": "Alabama",
5    "period": "2011",
6    "all_providers": "9676",
7    "all_primary_care_providers": "4064",
8    "all_physicians": "8281",
9
10
11    "State": "Alabama",
12    "Year": "2011",
13    "Health-Direct Expend": "$683,391",
14    "Health-Capital Outlay": "$13,697"
15 }
```

Methods

Nodes: Each state is a node | **Edges:** Two types of similarity connections between states:

- Provider Similarity (red edges) → From cosine similarity of provider metrics:
 - All providers, Primary care providers, Physicians, Nurse practitioners, Physician assistants
 -
- Spending Similarity (blue edges) → From cosine similarity of spending metrics:
 - Health & Hospital Direct Expenditure, Health Direct Expenditure

Edge Weights:

- Cosine similarity between normalized state features
 - Accounts for proportional similarities rather than absolute differences
 - Helps identify states with similar healthcare distribution patterns regardless of scale
- Threshold of 0.85 (85% similarity in patterns)
- Color gradient to show stronger similarities

Undirected MultiGraph | Provider similarity | Spending Similarity | Regions Color Coded

50 Nodes (States) | 427 Edges | 397 Edges | Total edges: 824

Healthcare Network: State Similarities by Region
Red = Provider Similarity, Blue = Spending Similarity

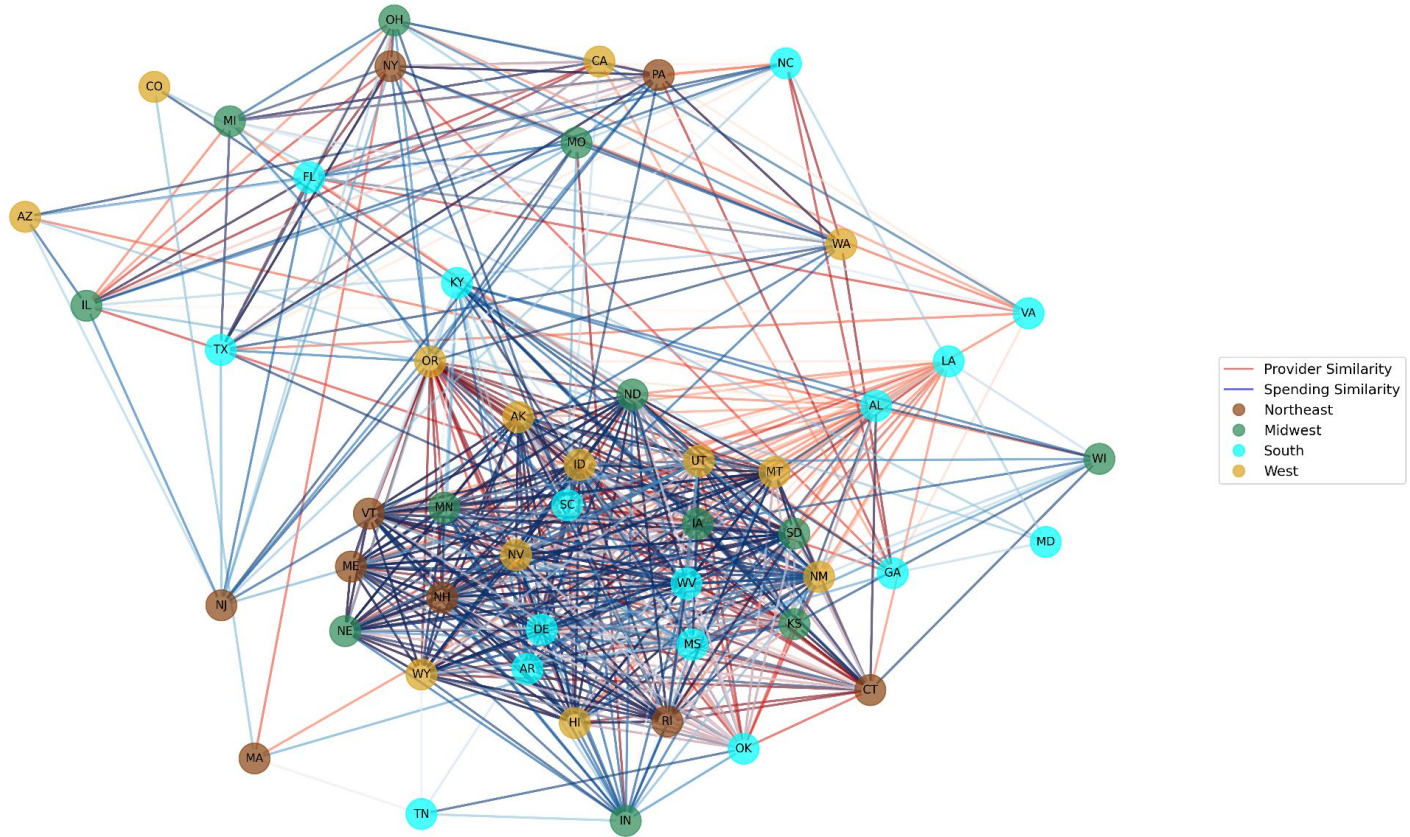


Table 2 States organized by BLS-defined region and the change in physician-related factors for 2017–2030

Rank	States	Region	Change in Population	2030 PHY Surplus	2030 PHY Shortage	2017 Shortage Ratio	2030 Shortage Ratio	2030 Ratios Surplus Shortage	2017 Grade	2030 Grade	Change in Shortage Ratios
1	Massachusetts	Northeast	212,993	10,133		-183	-145		A	A	39
2	Vermont	Northeast	31,138	673		-146	-95		A	B	52
3	New York	Northeast	-91,319	14,875		-105	-76		B	B	29
4	Rhode Island	Northeast	6,347	677		-93	-59		B	B	34
5	Connecticut	Northeast	34,457	1,409		-71	-38		B	C+	33
6	Delaware	South	69,907	334		-87	-33		B	C+	54
7	Maryland	South	695,261	1,408		-51	-20		C+	C	31
8	Nebraska	Midwest	25,380	274		-51	-15		C+	C	36
9	Ohio	Midwest	-94,920	1,648		-43	-14		C+	C	29
10	Maine	Northeast	12,607	175		-63	-12		B	C	51
11	New Hampshire	Northeast	161,873	161		-55	-10		C+	C	45
12	New Jersey	Northeast	459,502	557		-37	-6		C+	C	32
13	Michigan	Midwest	46,320	247		-32	-2		C+	C	30
14	Pennsylvania	Northeast	19,207		32	-32	0		C+	C	33
15	Indiana	Midwest	246,614		265	-24	4		C	C	28
16	Oklahoma	South	222,552		562	-17	14		C	C	31
17	Minnesota	Midwest	541,994		2,260	2	33		C	C-	31
18	Wisconsin	Midwest	214,335		2,263	-3	37		C	C-	40
19	North Dakota	Midwest	-27,067		234	-8	39		C	C-	47
20	Virginia	South	1,176,686		3,911	10	40		C	C-	30
21	Utah	West	622,179		1,418	20	41		C	C-	21
22	Oregon	West	724,952		2,008	19	42		C	C-	23
23	Colorado	West	652,453		2,424	15	42		C	C-	27
24	Illinois	Midwest	274,228		6,203	19	46		C	C-	27

26	Missouri	Midwest	306,012		3,102	18	48			C	C-	30
27	Iowa	Midwest	-71,659		1,660	19	56			C	C-	37
28	Hawaii	West	68,788		876	26	60			C	D	34
29	Texas	South	5,926,674		20,420	43	61			C-	D	19
30	Alabama	South	184,833		2,988	32	61			C-	D	29
31	West Virginia	South	-95,863		1,079	22	63			C	D	41
32	South Carolina	South	432,367		3,230	23	63			C	D	40
33	North Carolina	South	1,942,187		7,725	38	63			C-	D	25
34	Wyoming	West	-6,867		335	7	64			C	D	57
35	Kentucky	South	173,091		2,926	34	64			C-	D	31
36	Georgia	South	1,538,674		8,012	41	67			C-	D	25
37	Washington	West	1,490,025		6,037	54	70			C-	D	16
38	Alaska	West	118,729		609	18	70			C	D	52
39	California	West	5,484,918		32,669	49	70			C-	D	21
40	Arkansas	South	234,482		2,303	42	71			C-	D	29
41	Kansas	Midwest	70,905		2,102	39	72			C-	D	33
42	Florida	South	6,631,870		21,978	26	77			C	D	51
43	Arizona	West	2,845,080		8,280	38	77			C-	D	39
44	Tennessee	South	768,162		5,989	55	81			C-	D	26
45	Montana	West	35,093		894	33	86			C-	D	53
46	Idaho	West	294,759		1,743	64	88			D	D	24
47	Nevada	West	1,068,738		4,177	81	98			D	D	17
48	Louisiana	South	109,220		4,820	69	100			D	D	31
49	New Mexico	West	38,384		2,118	44	101			C-	D	57
50	Mississippi	South	64,467		3,709	85	120			D	F	35
All 50 States			35,891,599	32,571	171,731							
2030 National Net Physician Shortage				139,160								

Summary Statistics

Healthcare Provider Statistics Summary

Metric	All Providers	Primary Care	Physicians	Nurse Practitioners	Physician Assistants
Mean	14530.0	6124.0	12023.0	1465.0	1042.0
Median	10231.0	4230.0	8335.0	1159.0	667.0
Std Dev	15272.0	6285.0	13069.0	1258.0	1090.0
Min	1299.0	626.0	968.0	155.0	80.0
Max	77403.0	32540.0	67349.0	5132.0	4923.0
CV (%)	105.1	102.6	108.7	85.9	104.7
Skewness	2.16	2.19	2.25	1.43	1.87
Kurtosis	5.15	5.45	5.7	1.59	2.93

Summary Statistics

Healthcare Spending Statistics Summary

Metric	Hospital Direct Expenditure (thousands \$)	Health Direct Expenditure (thousands \$)
Mean	2495.0	996.0
Median	1896.0	593.0
Std Dev	2536.0	920.0
Min	154.0	94.0
Max	11791.0	4443.0
CV (%)	101.6	92.4
Skewness	1.89	1.65
Kurtosis	3.73	2.86

Results

Provider Distribution

- Major disparities: All Providers (Max: 77,403 vs Median: 10,231)
- Nurse Practitioners: Most “uniform” (CVar: 85.9%)
- Physicians: Most variable (CVar: 108.7%)

Spending Patterns

- Hospital expenditure: High variability (\$154K to \$11.8M)
- Health direct expenditure: More consistent (CVar: 92.4%)

Regional Connections:

- West: Most similar (66 connections/13 states)
- South: Large but diverse (55 connections/16 states)
- Midwest: Moderate (42 connections/12 states)
- Northeast: Most diverse (21 connections/9 states)

Key Findings

- High provider variability suggests uneven healthcare access
- Similar spending doesn't always indicate similar provider distribution

Problems + Solutions

Problems

- Very small population size, only covering 2011-2013
 - May lead to future prediction study
- Outdated data might not reflect current healthcare landscape
- Can't establish causation between spending, provider patterns, or how this even affects patient outcomes

Potential Solutions:

- Gather more data (tedious but possible, looking through US BLS)
- Adjusting thresholds on edge weights
- Add geographic distance as a factor (normalize this for rural vs urban populations)